Rossmoyne Senior High School

Semester One Examination, 2022

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two: Calculator-assumed

WA student number: In figures

In words

Your name

Circle your Teacher's Surname: Adams Bartoshefski Fletcher Pisano Vonk Younger

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Number of additional answer booklets used (if applicable):

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

SN085-193-6

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen.
Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
8	11	
9	3	
10	8	
11	8	
12	9	
13	8	
14	7	
15	8	
16	10	
17	8	
18	9	
19	9	
S2 Total	98	
S2 Wt (×0.6633)	65%	

Section Two: Calculator-assumed **65% (98 Marks)**

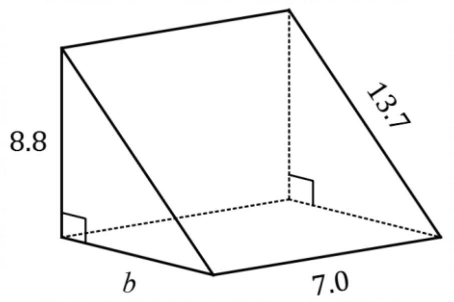
This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 8 **(11 marks)**

The diagram, not to scale, shows a right triangular prism with a length of 7.0 cm.

The lengths of some of the edges are shown on the diagram in centimetres.



(a) Determine b, the length of the base of the right triangle. (2 marks)

(b) Determine the total surface area of the prism. (4 marks)

End of questions

c) A carpenter wants to make this triangular prism out of solid wood and use jarrah, a local hardwood.
The density of jarrah is 835 kg/m^3.

Calculate the weight of the triangular prism when made out of jarrah.
Give your answer rounded to the nearest gram. (5 marks)

Question 9 **(3 marks)**

A 2.5Kg packet of dog biscuits was on sale for $11.20 from a veterinary shop.
A 1.5kg packet of the same dog biscuits cost $6.25 from a supermarket.

a) Which is the better buy, and by how much.

Question 10 **(8 marks)**

At the start of this year, an art supplies wholesaler in Australia needed to buy 3000 artist brushes and collected the following price information from overseas manufacturers:

Country of manufacturer	Hong Kong	Canada	Thailand
Number of artist brushes per pack	500	200	300
Cost per pack (in local currency, includes shipping)	$\text{HK\$}6850$	$\text{C\$}460$	$17\,000$ baht

At the time, the relevant exchange rates for one Australian dollar were 5.495 Hong Kong dollars, 0.898 Canadian dollars and 23.35 Thai baht.

(a) Determine the total cost in Australian dollars of buying 3000 brushes from the Canadian manufacturer. (3 marks)

(b) Determine the cost per brush from the Thai manufacturer in Australian dollars. (2 marks)

(c) Determine, with justification, which of the three countries the wholesaler should use in order to minimise the unit cost per brush. (3 marks)

End of questions

Question 11 **(8 marks)**

Information about the shares of three companies listed on the Australian Securities Exchange (ASX) towards the end of last year is shown in the table below.

ASX Share Code	Market value of share ($)	Earnings per share ($)	Dividend per share ($)	Percentage dividend
NST	9.47	1.046	0.18	$\boldsymbol{A}$
RHC	70.45	1.804	1.55	2.20%
GMG	26.09	1.221	$\boldsymbol{B}$	1.15%

(a) Determine the value of $\boldsymbol{A}$ and the value of $\boldsymbol{B}$ in the table. (2 marks)

(b) Calculate the price-to-earnings ratio for each share and hence rank the stocks in order from lowest to highest price relative to earnings. (3 marks)

(c) Calculate, to the nearest dollar, the total dividend paid to an investor who owned 915 shares in NST, 230 shares in RHC and 412 shares in GMG. (3 marks)

Question 12 **(9 marks)**

Ash lives at home with her parents. Her take-home income and all the expenses used in her personal budget from the previous year are shown below, together with how often the amounts are incurred and updates that Ash needs to make for the current year.

Use 1 year = 52 weeks and 1 year = 26 fortnight throughout this question.

Budget item	Previous amount ($)	Frequency	Update
Take-home income	2100.00	Fortnightly	Increase by 3.5%
Gym membership	22.00	Weekly	Increase by $3.50
Household contribution	140.00	Weekly	Increase by $30
Mobile phone	45.25	Monthly	No change
Car running costs	150.00	Weekly	Decrease by 10%
Car loan repayment	506.75	Monthly	No change
Entertainment & eating out	150.00	Weekly	Increase by 5%
Other	450.00	Monthly	Decrease by 5%

(a) Determine the amount of her fortnightly take-home income for the current year, after the change indicated in the update column. (2 marks)

(b) Use the information in the update column to complete the missing entries in the table below for Ash's budget for the whole of this current year. (3 marks)

Budget item	Amount this year ($)	Frequency
Take-home income		Fortnightly
Gym membership		Weekly
Household contribution		Weekly
Mobile phone	45.25	Monthly
Car running costs		Weekly
Car loan repayment	506.75	Monthly
Entertainment & eating out		Weekly
Other	427.50	Monthly

End of questions

After subtracting her expenses from her take-home income, Ash calculated that at the end of last year she was left with a surplus of $\$18\ 552$ to use for a holiday or deposit in a savings account.

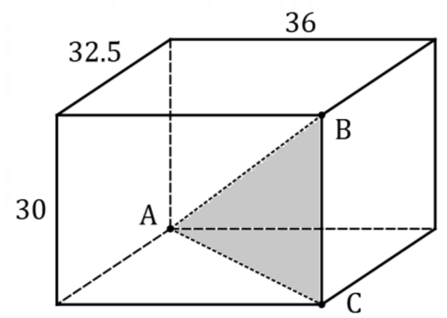
(c) Ash hopes that her budget updates for the current year will lead to an increase in her annual surplus. Determine, with justification, whether this is the case. (4 marks)

Question 13 **(8 marks)**

Six decorative spheres are packed into a cuboid shaped box. Each sphere has a diameter of 14 cm, and the internal dimensions of the box are 36 cm by 32.5 cm by 30 cm.

(a) Determine the percentage of the space inside the box that the spheres occupy. (4 marks)

(b) A thin triangular piece of card ABC is inserted into the box to limit the movement of the spheres, as shown in the diagram.



Determine the perimeter of the right triangle ABC. (4 marks)

End of questions

Question 14 **(7 marks)**

The table below and matrix $\mathbf{N}$ show the number of trips **(in thousands)** made in 2006 by visitors to three Australian cities categorised by reason (VR;visiting relatives or Holiday).

$(000's)$	VR	Holiday
Perth	1125	1060
Brisbane	1827	1544
Darwin	91	313

$$\mathbf{R} = \begin{bmatrix} 1.02 & 0 & 0 \\ 0 & 1.045 & 0 \\ 0 & 0 & 1.03 \end{bmatrix}, \quad \mathbf{N} = \begin{bmatrix} 1125 & 1060 \\ 1827 & 1544 \\ 91 & 313 \end{bmatrix}$$

Visitor numbers (in both categories) to Perth, Brisbane and Darwin increased by 2%, 4.5% and 3% per year respectively over the following years and this information is shown in matrix $\mathbf{R}$.

(a) Determine matrix $\mathbf{R} \times \mathbf{N}$, rounding entries to the nearest whole number. (2 marks)

(b) How many more VR trips were made to Brisbane in 2007 compared to the same type of trip to Perth in 2007? (2 marks)

(c) The matrix $\mathbf{R}^n \times \mathbf{N}$ will show the number of trips (in thousands) by category in the n^{th} year after 2006. Determine, with justification, the year in which the number of holiday trips to Brisbane first exceeded $2\ 000\ 000$. (3 marks)

Question 15 **(8 marks)**

The image shows some recently harvested grain stored in a conical pile on level ground.

The top of the pile is 12 m above the ground and the radius of the base is 35 m.

(a) Determine the volume of grain contained in the pile. (2 marks)

(b) Trucks that can carry a load of 28 tonnes will be used to transport the grain to the local port. If one cubic metre of grain weighs 630 kg, determine the number of truck loads of grain required to move the entire pile of grain to the port. Note that there are 1000 kg in one tonne. (2 marks)

(c) Determine the slant height of the conical pile. (2 marks)

(d) When rain is forecast, the sloping surface of the grain can be covered with tarpaulins. Calculate the area of grain that needs to be covered. (2 marks)

End of questions

Question 16 **(10 marks)**

(a) Kelly takes a short-term loan of $\$2400$ at 9.6% per annum simple interest for 50 days. Determine the sum of the principal and interest that Kelly must repay at the end of the loan period. (3 marks)

(b) An amount of $\$15\ 000$ is invested at 4.5% per annum compounded annually for 3 years. Determine the total interest earned on this investment over the 3 years. (3 marks)

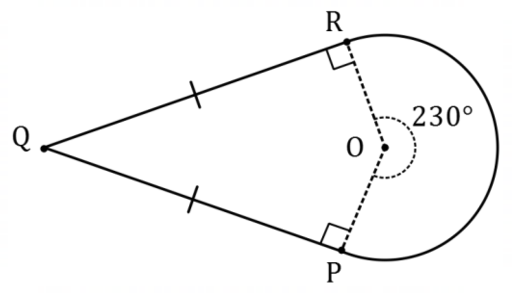
(c) At the start of last year, Syed was employed as a manager with a salary of $\$86\ 000$. During the year, Syed was awarded a 6.5% pay rise followed by another rise of 7.7% at the start of this year.

(i) Determine Syed's salary at the start of this year. (2 marks)

(ii) Determine the percentage increase in Syed's salary from the start of last year to the start of this year. (2 marks)

Question 17 **(8 marks)**

The diagram (not to scale) shows a fabric panel used in the construction of a kite.



The panel consists of a $230°$ sector of a circle with centre O and radius 50 cm, and quadrilateral $OPQR$ in which $QR = QP = 108$ cm.

(a) Calculate the area of the **sector**. (2 marks)

(b) Determine the **total area** of the fabric panel. (3 marks)

(c) The panel is to be edged with tape. Calculate the length of tape required. (3 marks)

End of questions

Question 18 **(9 marks)**

Covid created a travel bubble in 2021 where many people from both Australia and New Zealand would travel only across the Tasman.
Mary changed some Australian dollars ($AUD) into New Zealand dollars($NZD).
The exchange rate was given as $1AUD = $1.07NZD.
She exchanged $2000 AUD.

(a) How many $NZD did she receive? (1 mark)

(b) After spending most of her $NZD, she had $300NZD left.
What was the value of this in $AUD? (2 marks)

(c) Determine the conversion rate of $1 NZD to $AUD (2 marks)

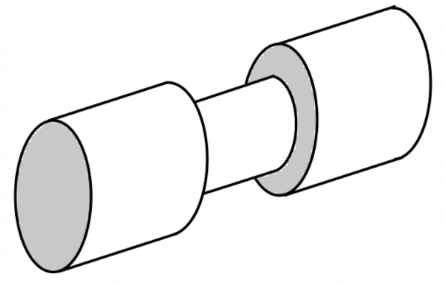
Mary's child had three notes; a $50, a $20 and a $10, but she didn't know which were NZ and which were Australian.

(d) Using all three notes, what was the **maximum** value possible in Australian dollars, if one of the notes was from NZ? (2 marks)

(e) Using all three notes, what was the **minimum** value possible in New Zealand dollars, if one of the notes was from Australia? (2 marks)

Question 19 **(9 marks)**

A woodturner takes a solid cylindrical wooden pole of length 66 cm and radius 8 cm and uses a machine to reduce the radius of the middle third of the pole by 30%, as shown in the diagram (not to scale).



(a) Calculate, with working, the volume of wood removed by the machine.

(4 marks)

End of questions

(b) Explaining your method, determine the total surface area of the remaining object.

(5 marks)

Supplementary page

Question number: _________

Supplementary page

Question number: __________